

with HP's parallel computing software, with optimised compilers and specialised math libraries.

Putting them through the paces

Over the past few decades, supercomputing muscle was exclusively used to run applications of strategic importance. Governments with nuclear arsenals use supercomputers extensively to simulate the testing of aging nuclear warheads, as well as to refine new designs in laboratory conditions. Other classified applications include solving computationally tough problems, such as cracking secret codes, and running warfare simulation exercises.

Research by quasi-governmental institutions into physics, mathematics and life sciences is something that also borrows a lot of supercomputing time. The NEC Earth Simulator (all of 35.8 TFlops), located in Japan, studies the earth and analyses atmospheric and oceanographic data.

Research in geosciences, such as seismic data processing, involving the processing of large data sets to simulate and analyse seismic behaviour; long term weather forecasting that involves running

numbers of variables over large data sets; computing protein compounds to researching new cures; these are all applications of supercomputing.

What's changed since the end of the cold war? With the emergence of cheaper cluster supercomputing, a lot of supercomputing cycles are now being bought and put to use by corporations worldwide. Daimler-Chrysler—the car manufacturer, for instance, uses extensive supercomputing power through their HP Superdome machines, as do many companies in the automobile and aerospace industry. Whether it's designing steering wheels, complex aircraft sections, crash testing, or even prototype development, most of it is done pretty accurately on supercomputers.

Another big customer is the finance industry. World-wide financial transactions are processed on large HPC machines—right from processing a credit card bill to scanning behavioural trends across GDPs, population data and so on. The NASDAQ, and other financial bourses of the world, depend extensively on the reliability and computational speed of supercomputers.

A recent emerging customer base for raw computational power is the film industry; the FX (special effects) in recent movies come courtesy of such computers. In fact, animated movies depend extensively on such machines; *Shrek*, for instance, was made by the DreamWorks studio in partnership with HP, using their HPC machines.

Shared Iron leads

The supercomputing trends of the future indicate that supercomputers of tomorrow will work faster, solve even more complex problems, and provide wider access to supercomputing power. However, another challenge in the making is getting supercomputer horsepower in smaller, more efficient packages. Today's supercomputers take up huge space—around 2000 sq ft and more—with miles of cabling and consume enough power to light up a small town. Tomorrow's supercomputers, may well take up the challenge of giving you all the computing power you need and more whilst using as much resources as a desktop. 

SRINIVASAN RAMAKRISHNAN
srinivasan_ramakrishnan@thinkdigit.com

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